

Alexander Bloomer

424-327-7445 | bloomer@calpoly.edu | <https://www.linkedin.com/in/alexander-bloomer>

EDUCATION

California Polytechnic State University, San Luis Obispo

Bachelor of Science, Computer Science

GPA: 4.0 Expected June 2028

Relevant Coursework: CSC 203: Project-Based Object-Oriented Programming and Design (Java), CPE 225: Introduction to Computer Organization (C, Assembly), Calculus 3, Linear Analysis

Redondo Union High School, Redondo Beach, CA

High School Diploma

GPA: 4.5

Relevant Coursework: AP Computer Science A, AP Computer Science Principles, AP Calculus AB, AP Calculus BC, AP Physics 1

El Camino College, Torrance, CA

No Degree

GPA: 3.75

Relevant Coursework: Computer Science 1: Problem Solving and Program Design using C++, Computer Science 2: Introduction to Data Structures (C++)

EXPERIENCE

The Coder School

Code Coach, March 2023 – Present

- Taught students from 6 to 18 to code in various languages and frameworks including Python, JavaScript, C#, Java, C++, CSS, HTML, Scratch, and Unity
- Engaged students with interesting programming projects to foster creativity and problem-solving skills.
- Led summer and winter Camps of up to 15 students on topics like scratch, robotics, and animation

Beach Cities Robotics, First Robotics Competition

Lead Technician, August 2021 - August 2022

Programming Student, August 2020 – June 2024

Co-Driver, August 2022- June 2024

- Collaborated with team members to design, build, and program several robots.
- Applied programming concepts to control robots using Java.
- Led setup and troubleshooting of the robot as lead technician.
- Cooperated with another driver to operate the robot in competitions, winning a regional competition, a World Championships division Championship, and reaching World Championship finals.
- Won Excellence in Engineering Award and Entrepreneurship Award.

Boeing

Software Engineering Intern, June 2023– August 2023

- Developed a robust pipeline for ensuring safety and security in code using Docker containers.
- Collaborated with Boeing software engineers to address complex pipeline challenges
- Presented pipeline to employees and chief engineers.
- Collaborated with three other interns to present innovative AI strategy to chief engineers and won an AI innovation award.

Cal Poly Robotics Club

Member, September 2024 – present

- Designed and developed Django-based website to remotely control a robotic boat.
- Collaborated with small team to optimize functionality and usability
- Used Agile development process to enhance teamwork and improve efficiency.

CyberPatriot

Team Member, August 2021 – present

- Collaborated with a 5-person team to secure computer systems by mitigating security threats.
- Developed automation scripts to expedite the process and improve the efficiency of risk detection.
- Ranked top 20% in the nation and first in school in national competitions.

PERSONAL PROJECTS

Baseball Simulator

- Developed a Python-based baseball simulator with Flask web interface
- Processed game outcomes using CSV data with real player statistics from Baseball Reference.
- Implemented probability-based calculations for batting outcomes based on player's stats (single, walk, out, etc.) and managerial decisions (e.g. pulling a pitcher).
- Designed understandable and interactive UI using HTML, CSS, and JavaScript, allowing users to easily simulate games or seasons and view results.
- Generated detailed game and season results for team analysis.
- Published site using Render.

Personal Website

- Designed and built a multi-page personal website using HTML and CSS
- Used best programming practices for version control with Git and GitHub

Chrome Extension for Video Speed Control

- Created a Chrome extension that allows users to speed up or slow down videos with custom button clicks.
- Implemented JavaScript to modify video playback speed dynamically.

TECHNICAL SKILLS

Programming Languages: Python, Java, C++, C#, C, RISC-V Assembly

Software Tools and Frameworks: Unity, Microsoft Excel, Git, GitHub, VScode, IntelliJ, Docker, Bash

Web Development: Django, Flask, HTML, CSS, JavaScript

HONORS

Ivy Chain Award – Top 50 juniors by GPA (2023)

AP Scholar With Distinction (2023)

California Scholarship Federation (2020-2023)

Robotics Programming Student of the Year (2023)

Robotics Rookie of the Year (2021)

Dean's List Honors (2024)